

Currently what sectors do you see students gravitating towards the most?

Students are gravitating the most towards AI, data science, software engineering architecture and development tools, security, machine learning, deep learning, VLSI design, networking and communication.

How does IIIT Bangalore enable startups?

We provide our students and faculty members with a seed fund for startups. We have a dedicated incubation centre that helps them work on their startup. We have accelerator programs to enable startups incubated under IIIT Bangalore and highly trained faculty also help most of the startups technically as mentors. We also have technology leaders visit the campus to interact with these startups.

How do you counter the industry-academia gap to make more of the graduate engineers employable in the VLSI design industry?

We have many Indian startups, SMEs and VLSI design companies that have come up and are scaling their operations. They will absorb many of these engineering graduates. Except for a few top institutes, such as the IITs, IIITs, NITs, BITS etc, which are teaching VLSI design and encouraging research, the knowledge of graduate engineers is limited.

The top companies need highly skilled manpower and that number is not so high in India. We have a lot of people at the ITI (Industrial Training Institute) level. So that kind of value chain has to be increased to highly skilled manpower, in niche technology areas. The number of engineers graduating from the top institutes is not very high and the industry is looking for people possessing that level of skill. The industry people are putting a lot of

requests to us to increase the number of highly skilled manpower we are working on.

How are you working on that?

For that, we are planning to make many expansions with respect to space and hiring highly skilled faculty members, and building better labs.

Do you believe that we have enough academicians who can aid in creating skilled engineers?

As a country, we need more good academicians who have an interest in teaching and research and have a good grasp of basic concepts. Good teaching and research is only possible when we have very creative and hardworking people.

How do we encourage people to gravitate towards academia?

We need a lot of policy changes and an infusion of a lot of resources to attract good faculty members, providing them with a lot of facilities for their research and teaching. So, these resources need to be handed over for that. Except for a few institutes, many institutes lack these facilities. Here, at IIIT Bangalore, we have most of the resources and facilities for our faculty.

What are the novel research projects and innovations that are currently under work at IIIT Bangalore?

We have been working on advanced cryptography, robotic surgery, data science, theoretical computer science, software testing, deep learning based image processing, 5G advanced and 6G technology, drone technology, digital governance, ethical IoT, and VLSI design, etc.

How do you see 5G technology impacting industrial IoT?

Very good question. Let me tell you, industrial IoT is a very large canvas and it has applications in

various areas. Industrial IoT, or any information technology, has three fundamental states. I call them the '3Cs'—Conservation of information, Computation of information, and Communication of information.

Conservation refers to storage. Any information in electronics is stored in the form of ones and zeros where one means high voltage and zero means low voltage.

The second 'C' is computation. After the conservation of information, we call the stored information to the CPUs and GPUs, and then we compute it according to our algorithms, such as an artificial intelligence (AI) or machine learning (ML) algorithm.

Once computed, we need to communicate this information. Industrial IoT (Internet of Things) has many sensors which collect data and compute it, and the data will be communicated to the receiver to take the required action.

5G is the last mile of communication that gives much better performance. If your communication is faster, then your performance will be better and thus more tasks can be automated, ultimately providing a boost to industrial IoT.

Wireless technologies consume a lot of power, so how feasible is it to expect these technologies to fit in the Indian scenario where power is still a huge problem?

I come from a networking background and I am aware of what you're talking about in remote/rural areas. We have solved this problem from 60% uptime to 99% uptime by using solar panels connected to the batteries in remote areas. These batteries are then connected to the communication devices. So our uptime is now 99%. Even with four hours of regular power in remote areas, we can continue work without any problem because of solar and other renewable energy. **EFY**